Simulation exercise for influenza A(H5N1) response on access, allocation, and deployment of medical countermeasures

meeting report
Geneva, Switzerland
4–5 September 2025

World Health Organization

Simulation exercise for influenza A(H5N1) response on access, allocation, and deployment of medical countermeasures

meeting report
Geneva, Switzerland
4–5 September 2025

World Health Organization

© World Health Organization 2026

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; **https://creativecommons.org/licenses/by-nc-sa/3.0/igo**).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: "This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition".

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization (**http://www.wipo.int/amc/en/mediation/rules/**).

Suggested citation. Simulation exercise for influenza A(H5N1) response on access, allocation, and deployment of medical countermeasures: meeting report, Geneva, Switzerland, 4-5 September 2025. Geneva: World Health Organization; 2026. **https://doi.org/10.2471/B09685**. Licence: **CC BY-NC-SA 3.0 IGO**.

Cataloguing-in-Publication (CIP) data. CIP data are available at **https://iris.who.int/**.

Sales, rights and licensing. To purchase WHO publications, see **https://www.who.int/publications/book-orders**. To submit requests for commercial use and queries on rights and licensing, see **https://www.who.int/copyright**.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers' products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

Contents

Acknowledgements **iv**
Abbreviations **v**
Executive Summary **vi**
1. Introduction **1**
2. Simulation overview **2**
2.1 Simulation scenario and injects 2
2.2 Participants 2
2.3 Methodology 3
3. Key insights from the simulation **4**
3.1 Information for decision-making 4
3.2 Regulatory pathways, policy recommendations, and legal considerations 5
3.3 Manufacturing, supply and logistics 7
3.4 Financing and incentives 9
3.5 Country readiness and impact on demand 10
3.6 Coordination, partnerships and urgent needs 11
4. Summary analysis and implications for access and allocation planning **14**
5. Way forward **16**
Annexes **17**
Annex 1. Agenda 17
Annex 2. List of participants 19

Acknowledgements

The World Health Organization (WHO) expresses its gratitude to the many individuals involved in the development of this meeting report.

The development of the report was coordinated by Ioana Ghiga who also lead the writing together with Alexandra Hill, Daniel Eibach and Shoshana Goldin (Epidemic and Pandemic Management Department, Medical Countermeasures Unit).

WHO would like to thank the participants and observes who attended the simulation exercise:

- WHO: Chikwe Ihekweazu (WHO Health Emergencies Programme), Maria Van Kerkhove (Department of Epidemic and Pandemic Management), Tim Nguyen (Medical Countermeasures Unit), Ioana Ghiga (Medical Countermeasures Unit), Alexandra Hill (Medical Countermeasures Unit), Daniel Eibach (Medical Countermeasures Unit), Razieh Ostad Ali Dehaghi (Regulation and Safety Unit), Razieh Ostad Ali Dehaghi (Regulation and Safety Unit), Alireza Khadem Broojerdi (Prequalification Unit), Wenqing Zhang (Global Respiratory Threats Unit), Anne Marie Huvos (Global Respiratory Threats Unit), Olga Kim (Global Respiratory Threats Unit), Kate Rawlings (Global Respiratory Threats Unit), Alexandra Kontic (Global Respiratory Threats Unit), Sandra Jackson (Global Respiratory Threats Unit), Lorenzo Subissi (Emerging Zoonotic Diseases Unit), Jamie Rylance (Clinical Management and Operations), Tara Prasad (Immunizatoin Decision-Making Unit), Georgios Stathopoulos (Immunization Decision-Making Unit), Alba Vilajeliu (Vaccine Research Development and Policy Unit), Francisco Nogareda (WHO Regional Office for the Americas), Marisol Valenzuela Lara (WHO Regional Office for the Americas), Pernille Jorgensen (WHO Regional Office for Europe), Emanuele Bruni (WHO Regional Office for Europe), Misheel Enkhdalai (WHO Regional Office for the Western Pacific), Pushpa Wijesinghe (WHO Regional Office for South-East Asia), Reena Hemendra (WHO Regional Office for Africa).
- Coalition for Epidemic Preparedness Innovations (CEPI): Saul Walker, Roshni Best
- Developing Countries Vaccine Manufacturers Network (DCVMN): Rajinder Kumar Suri
- Foundation for Innovative New Diagnostics (FIND): Emmanuel Agogo
- GAVI, the Vaccine Alliance: Hannah Kettler, Sophie Mathewson, Simon Allan
- Health Emergency Preparedness and Response (HERA): Luciana Da Silva Santos
- International Federation of Pharmaceutical Manufacturers and Associations (IFPMA): Paula Barbosa
- International Federation of Red Cross and Red Crescent Societies (IFRC): Gwendolene Eamer
- United Nations Children's Fund (UNICEF): Ann Ottosen, Siri Bjornstad, Hans Christiansen, Johannes Kraus, Nagwa Hasanin, Anu Puri, Maria Wilda Silva
- UNITAID: Carmen Perez Casas, Pearl Ngugi
- Medicines Patent Pool (MPP): Tiwadayo Braimoh

The document was formulated in accordance with the existing work plan under the implementation of the Pandemic Influenza Preparedness (PIP) Framework. WHO expresses its gratitude to the PIP Framework Partnership Contributions for supporting the development and publication of this document.

Abbreviations

GISRS	Global Influenza Surveillance and Response System
IVPP	Influenza Viruses with Pandemic Potential
MCMs	Medical Countermeasures
PIP	Pandemic Influenza Preparedness Framework
SMTA2	Standard Material Transfer Agreements 2
WHO	World Health Organization

Executive Summary

Influenza A(H5N1) and other influenza viruses with pandemic potential (IVPP) continue to pose a significant global public health threat. Ensuring timely, equitable, and effective access to medical countermeasures (MCMs)—including vaccines, antivirals, and diagnostics—requires strong global coordination, equitable allocation mechanisms, and robust operational readiness. In this context, WHO conducted a two-day simulation exercise to examine how operational assumptions, coordination arrangements, and allocation considerations might function during an emerging influenza A(H5N1) pandemic. The exercise provided an opportunity to explore how WHO, Member States, international organizations, regional bodies, and private-sector partners could work together to navigate evolving supply conditions and increasing demand for vaccines, antivirals, diagnostics, and other MCMs.

Through a phased scenario with escalating pressures, the exercise aimed to surface potential gaps and areas where further clarity or strengthening may be beneficial. The discussions also sought to identify ways to reinforce collaboration across levels of the system so that MCMs can be accessed, allocated, and used in an equitable and timely manner during a real-world outbreak.

A consistent theme across the exercise was the interdependence of systems that inform and enable allocation decisions. Rather than viewing allocation as a discrete step, participants emphasized the need to consider how surveillance, regulatory pathways, supply and manufacturing dynamics, operational readiness, financing, and country-level capacities collectively shape what is feasible and appropriate at any given moment. The diagnostic landscape, in particular, emerged as an area warranting further examination given its foundational role in informing both epidemiological understanding and subsequent decisions on vaccines and antivirals.

Participants noted that timely, reliable, and comprehensive information is central to supporting allocation considerations. Surveillance and laboratory data—including those generated through the WHO Global Influenza Surveillance and Response System (GISRS) and other established mechanisms—along with market, supply, and country readiness information were viewed as essential inputs. Increased visibility into available supply, bilateral arrangements, and stockpiles was seen as potentially supporting more informed and transparent prioritization. Modelling, risk assessments, and demand estimations were recognized as useful tools, while also requiring continuous updating as new information becomes available.

Regulatory readiness was highlighted as an important determinant of when and how MCMs may become available for use. WHO processes such as prequalification and the Emergency Use Listing, as well as reliance and recognition mechanisms, were viewed as supporting timely access. Participants noted that regulatory considerations may differ across product types, particularly for next-generation vaccines. Clear, coherent policy guidance—including on liability and indemnity—was identified as important for supporting timely access.

Manufacturing capacity, supply chain constraints, and delivery capabilities were recognized as key factors influencing both allocation decisions and implementation. Pandemic vaccine production remains concentrated within specific platforms and geographies, with implications for availability and timelines. Diagnostics were identified as an area where supply chain vulnerabilities—including dependencies on reagents and platform-specific components—could be pronounced. Delivery considerations, such as cold-chain capacity, infection prevention and control requirements, and the availability of ancillary supplies, were seen as important determinants of absorptive capacity for vaccines.

Participants noted that timely, flexible, and predictable financing is an important enabler of access to MCMs. Early funding—whether through donor envelopes, emergency mechanisms, or other instruments—can support procurement, manufacturing scale-up, and operational readiness. Risk-sharing arrangements and “no-regrets” approaches were discussed as potentially useful in contexts of early uncertainty.

The exercise highlighted that country-level preparedness plays a substantial role in determining the real-world impact of MCM allocations. Regulatory readiness, national deployment and vaccination plans, governance mechanisms, public acceptance, and immunization infrastructure were noted as relevant factors. Existing seasonal influenza programmes, where present, may provide valuable foundations. Demand estimates, participants suggested, should take into account behavioural, logistical, and operational considerations to ensure alignment between allocated supply and realistic uptake.

Strong coordination across WHO headquarters, Regional Offices, Member States, regional entities, and partners was identified as essential. Regional bodies were recognized as increasingly relevant actors, particularly when well connected to global processes. Participants discussed the value of clear communication channels and early triggers for engagement, observing that reliance solely on formal declarations, such as a public health emergency of international concern, may not always enable the earliest or most appropriate actions. Considerations regarding humanitarian settings were also raised, indicating the value of further exploring tailored approaches.

Insights from the simulation will inform ongoing refinement of the WHO Global Allocation Framework for pandemic influenza MCMs. Areas suggested for additional attention include strengthening diagnostic considerations, clarifying early triggers and information flows, improving visibility into supply and demand, considering roles and responsibilities, exploring flexible financing arrangements, and addressing needs in humanitarian contexts. Continued focus on country readiness and practical implementation capacities must also remain central to advancing timely, equitable, and effective access in a future influenza pandemic.

1. Introduction

A(H5N1) avian influenza along with other influenza viruses with pandemic potential (IVPP), remains significant global public health threat. Effective response to such a threat depends not only on early detection and the availability of medical countermeasures (MCMs), including vaccines, antivirals, and diagnostics and other MCMs, but also on their timely, equitable, and efficient upstream access and allocation, and effective collaboration and coordination among partners and stakeholders. Mindful of this, WHO undertook a two-day simulation exercise with the aim of testing operational assumptions, coordination approaches, and allocation considerations.

The exercise sought participation from WHO headquarters, WHO Regional Offices, Pandemic Influenza Preparedness (PIP) stakeholders – in particular industry associations and civil society, regional bodies, United Nations and international partner organizations. It also examined the interactions between triggers, access, allocation and deployment, highlighting how production, stockpiling and visibility of advance purchase agreements, and national capacities for roll-out to name a few, influence the equitable distribution of MCMs during a pandemic scenario.

Key global structures underpinning this process include the WHO Global Influenza Surveillance and Response System (GISRS). GISRS provides critical epidemiological surveillance data, real-time virus detection and characterization, reference viruses for diagnostics, antiviral susceptibility, candidate vaccine viruses and candidate vaccine viruses for H5 and other IVPP for influenza pandemic preparedness. This information guides MCMs development and prioritization, along with the PIP Framework, which provides WHO with the mandate to co-ordinate influenza pandemic preparedness and response as well as ensures legally binding commitments for equitable access to vaccines, antivirals, and diagnostics through instruments such as Standard Material Transfer Agreements 2 (SMTA2). The exercise also leveraged insights from the i-MCM-Net initiative, which fosters collaboration across global and regional networks to align operational capacities and support timely allocation decisions. The simulation exercise considered a hypothetical influenza A(H5N1) pandemic scenario, providing insights into:

- Enhancing coordination approaches among WHO, Member States, and partners—including international organizations, regional bodies, and private sector stakeholders—in relation to allocation strategies.
- Identifying gaps and areas for improvement in the supply and allocation of vaccines, antivirals, diagnostics, and other MCMs.
- Strengthening collaboration and communication pathways to ensure the equitable and timely access, allocation, and uptake of critical resources during an outbreak.

This simulation exercise recognized that informed decision-making requires data, evidence, and communication. It was designed to identify actions and activities that should be conducted during the interpandemic and emergence periods to support rapid response despite the uncertainties of the pandemic emergence.

This report documents the discussions, observations, and lessons learned during the simulation. It highlights practical considerations for improving global readiness and provides considerations to inform the development of the global allocation framework and its operational implementation for future influenza pandemics.

2. Simulation overview

2.1 Simulation scenario and injects

The simulation exercise centered on a hypothetical A(H5N1) avian influenza outbreak that escalated progressively into a pandemic scenario. The scenario was structured to replicate real-world outbreak dynamics, including detection, initial response efforts, virus characterization, and the global demand for vaccines, antivirals, and diagnostics.

The exercise included a series of scenario injects designed to prompt considerations needed for decision-making and coordination at multiple levels. The initial detection and confirmation phase began with early reports of suspected influenza A(H5N1) cases in multiple countries. GISRS provides year-round surveillance enabling early detection and virus characterization of IVPP, highlighting its importance as a global alert system for timely detection and its implication for the efficient deployment of antivirals and other MCMs (e.g. personal protective equipment) to contain the outbreak.

During the regional spread phase, evidence of transmission across neighboring countries emerged, requiring participants to consider cross-border coordination. This stage generated requests for access to MCM that may be available through different arrangements such as stockpiles (or contractual arrangement for access to surge supplies), and country readiness for the deployment of MCMs to affected regions.

The pandemic emergency phase followed, triggering an escalation in demand for pandemic influenza vaccines, diagnostics, and antiviral treatments. The pandemic emergency determination resulted in the activation of the PIP Framework mechanisms, including SMTA2 commitments.

Finally, operational challenges and decision points were introduced to understand perspectives and actions to respond under different challenges. These included delays in manufacturing scale-up and logistics, prioritization of high-risk populations, allocation across regions with varying capacities, communication and coordination challenges, and other pressures that have been documented in the past when dealing with allocations of scarce resources.

2.2 Participants

The exercise engaged a broad range of stakeholders across different levels, reflecting the multi-sectoral coordination required in a pandemic response.

A detailed List of Participants is in Annex 2.

2.3 Methodology

The simulation employed a multi-layered methodology designed to capture both operational decision-making and reflective learning. Progressive scenario injects prompted participants to make real-time decisions on access, allocation, and prioritization of MCMs reflecting the complexities of an evolving outbreak. Following each inject, participants engaged in structured discussions to examine challenges, bottlenecks, and coordination strategies, with debriefs emphasizing operational lessons learned and areas for improvement. Finally, participants undertook structured reflection and analysis at the end of the exercise (a hotwash) to capture insights on overall preparedness, gaps that need to be urgently addressed and way forward.

This methodology allowed the exercise to mirror the complexity of real-world pandemic response, ensuring that lessons learned are directly relevant to strengthening access and allocation strategies for vaccines, antivirals, and diagnostics.

3. Key insights from the simulation

The simulation exercise highlighted the complex interplay of factors that influence access and allocation of MCMs. This chapter organizes key observations and insights aiming to provide a holistic perspective on the enablers, bottlenecks, and considerations for strengthening global preparedness and response.

3.1 Information for decision-making

Accurate and timely information was anticipated to form the cornerstone of effective decision-making in the simulation, directly influencing equitable access and allocation of MCMs. The GISRS - with its various databases of RespiMART (virologic -FluNet and epidemiological – FluID) and related genomic sequencing sharing were positioned as essential for detecting emerging influenza A(H5N1) cases, characterizing the virus, epidemiological trends and characteristics of human cases and risk assessments. Data on circulating viruses from GISRS was expected to play a central role in supporting the timely development and early deployment of diagnostics, antivirals, and vaccines underscoring the importance of integrated global and national surveillance networks.

Early outbreak: situational awareness and preparedness inputs

In the initial phase of the outbreak, much of the focus would be on building a picture of the epidemiological situation the global MCM landscape of licensed and pipeline products and supporting active responses in affected countries. Enhanced diagnostic and surveillance outputs and early WHO risk assessments would provide an initial sense of the severity of the strain, while the Influenza Investigations and Studies Global Initiative, also known as the Unity Studies insights, could help assess transmissibility, immunity, and infection/disease severity. In parallel, preparedness activities such as WHO's production capacity assessments for vaccines and antivirals, as well as the INTREPID Alliance therapeutics landscape review, were recognized as important foundations. Market analysis was expected to extend beyond product pipelines and production capacity to also include regulatory approvals, supply chain resilience, and intellectual property arrangements, especially for products with long production cycles.

Escalation and response: demand forecasting and operational intelligence

As the scenario progressed toward regional spread and determination of pandemic emergency under the IHR, information needs were anticipated to shift toward demand forecasting and operational intelligence. Research evidence, epidemiologic modeling, and risk assessments would become critical for guiding strategy and policy guidance and allocation priorities, while the WHO Global Clinical Platform was highlighted as a tool for real-time clinical data sharing and visualization under a robust legal framework. Participants anticipated the value of interoperable, federated-data approaches — warning that multiple fragmented dashboards could create confusion, whereas a small number of authoritative platforms would allow more coherent decisions. Procurement of medical countermeasures should be coordinated to ensure equitable and timely access. Visibility on bilateral deals and donations as well as on national stockpiles and stock levels was also recognized as a critical

informational gap: without this transparency, risks of inequitable oversupply, under or overestimation of demand, or undersupply across countries could increase. Data sharing can support planning and resource allocation.

Post-allocation: feedback and adjustment

Needs for accurate information will persist after initial allocation decisions. Monitoring demand, uptake, and residual stock would provide essential feedback for subsequent rounds of allocation, helping align resources with country readiness and demand. This ongoing flow of information was seen as central to ensuring both equity and efficiency over the course of a pandemic response.

Summary of key aspects

- Surveillance and risk characterization through GISRS, Unity Studies, and WHO risk assessments would anchor early action and inform strategy and policy setting.
- Market analysis — including production capacity, supply chains, regulatory approvals, and intellectual property arrangements — would complement epidemiologic intelligence.
- Forecasting and modeling would help guide prioritization, but their utility depends on data availability and quality.
- Information sharing related to bilateral deals and donations as well as stockpiles and residual stocks would avoid mis-estimations in available supply.
- Integration of systems through interoperable platforms would build coherence and trust in allocation decisions.
- Post-allocation monitoring of demand and absorption would help adjust and recalibrate future allocations and related distribution.
- It was reinforced that information is not a static input but a dynamic, evolving requirement throughout the outbreak trajectory and observed effectiveness of vaccines, potential resistance to antivirals, etc
- The allocation framework will be developed during the interpandemic and emergence periods; it will leverage learnings from previous epidemics and pandemic (including but not limited mpox, ebola, influenza, and COVID-19 experiences).
- Partner coordination should begin before a public health emergency of international concern is declared. During the event, partner activities will be coordinated to ensure the relevant entities can contribute their expertise and capacities, in accordance with their mandates. Preparedness activities will be undertaken to enable this communication and coordination.

3.2 Regulatory pathways, policy recommendations, and legal considerations

Regulatory and policy setting mechanisms (e.g. the WHO Strategic Advisory Group of Experts on Immunization, Clinical Guideline Development Group) were anticipated to act as critical enablers to inform allocation. The simulation highlighted the central role of national regulatory agencies, WHO prequalification and emergency use listing, and reliance mechanisms in determining how quickly diagnostics, antivirals, and vaccines could reach affected populations.

Setting clear public health goals, choosing specific strategies, and identifying priority groups—especially when supplies are limited, will be important to support country readiness decisions. National Deployment and Vaccination Plans were expected to serve as essential instruments to translate global guidance and provide a single source of timely information on national capacities

for MCM roll-out. Their relevance was particularly evident in the context of products made available through the PIP Framework and SMTA2 commitments (please see Annex for information on SMTA2 signatories). Through the PIP Framework, WHO is conducting legal negotiations in the interpandemic and emergence periods to support expedited decision-making and availability of supplies during an event.

Regulatory readiness and national regulatory agencies maturity levels

Discussions anticipated the need to understand regulatory capacities within the context of ongoing efforts of assessing maturity levels of national regulatory systems. For countries with MCM production capacity, this maturity would be especially important in shaping both domestic approvals and the quality assurance of products for export; a minimum maturity level 3 being required. Beyond initial approvals, capacities for pharmacovigilance and active post-introduction monitoring were seen as critical to maintaining trust in MCMs. Quality assurance emerged as a cross-cutting concern: understanding which products were circulating, and whether they met international standards. Clarity on the regulatory pathways for mRNA and other new vaccine platform vaccines is needed prior to a pandemic emergency; this is likely to be a critical factor for ensuring rapid access to next-generation influenza vaccines, supplementing the supply produced through traditional technologies. There is a need for clarity on potential correlates of protection to enable regulatory approval of next-generation influenza vaccines. Regulatory harmonization would also facilitate marketing authorization of next-generation influenza products.

Interaction of global and national frameworks

The scenario also emphasized the need for alignment between WHO guidance, national regulatory requirements, and emergency use procedures. From past experiences, where reliance mechanisms were weak or absent, delays could cascade into major bottlenecks. Participants anticipated that early experiences from countries that would begin introducing MCMs could provide valuable lessons, helping to refine global recommendations and national practices in real time. Furthermore, alignment of recommendations – e.g. between regulators and policy bodies (such as the WHO Strategic Advisory Group of Experts on Immunization) was also stressed.

Legal and risk considerations

Liability and indemnity schemes were anticipated to be among the most sensitive issues including. no-fault compensation schemes. Legal aspects related to allocations for humanitarian settings raised additional questions about feasibility, fairness, and the need for comprehensive liability mechanisms. The simulation also drew attention to the importance of clear, updated global guidance on donations. Concerns were raised, that in particular for donations, without enough visibility, ad hoc bilateral or regional arrangements may risk creating uncertainty, inequities, or duplication of effort.

Summary of key aspects

- Regulatory maturity matters: countries need capacity not only to authorize but also to monitor and ensure quality.
- The WHO prequalification and emergency use listing is a key step for global procurement and should be prioritized.
- Global, regional, and national policy setting bodies will play a key role to identify priority groups
- National Deployment and Vaccination Plans serve as a bridge between global commitments and national action, but their absence or weakness can slow access. Several countries are also opting to develop MCM-deployment plans s (deployment plans that consider all MCM and related activities) or develop very detailed pandemic plans that contain all these provisions.
- Experiences from early adopters could provide real-time evidence to strengthen evolving policy recommendations and deployment strategies.
- Liability, indemnity, and compensation remain unsettled issues, particularly in fragile or humanitarian settings.
- Clear, globally agreed donation frameworks would reduce uncertainty and foster fairness in allocation.

3.3 Manufacturing, supply and logistics

From WHO's analyses of production capacity, it was evident that existing production capacity for vaccines and antivirals remains heavily concentrated in high- and upper-middle-income countries, creating the potential for inequities in the availability of MCMs during a rapidly escalating outbreak. Diagnostic production capacity, which will impact the availability and access to diagnostics, were highlighted as an early bottleneck of the response to outbreaks: discussions emphasized the need to scale up different diagnostic assays production capacity in a timely manner.

Manufacturing capacity and timelines

As the scenario evolved toward regional spread, the risk of delays in scaling up production became more visible. While some antiviral stocks may already be available, discussions revealed that expanding production may face lead-time challenges, particularly around securing active pharmaceutical ingredients . Over 80% of seasonal and pandemic vaccine production capacity relies on embryonated eggs (in comparison, cell-based influenza vaccines comprise ~11% of seasonal influenza and ~16% of pandemic influenza production capacity). Inactivated influenza virus vaccines comprise the majority of the seasonal and pandemic influenza vaccine supply. Newer technologies, including but not limited to mRNA, are currently in the clinical pipeline. Following licensure, these technologies can offer greater flexibility but are anticipated to be constrained by limited capacity in the near term, unclear regulatory pathways, and potentially higher prices than vaccines produced with traditional technologies.

Participants noted that equitable and timely allocation would depend not only on the scale of production but also on how quickly products would be licensed and deployed.

Seasonal influenza vaccine production cycles were flagged as another factor that must be taken into account. Starting production of pandemic vaccines could entail challenges in switching from seasonal production which would largely depend on the stage of the manufacturing cycle, and the recognition of the existence of contractual agreements for seasonal influenza vaccine delivery for manufacturers (with potential room for flexibility).

In 2022, seasonal influenza vaccines are available in 143 countries (74% of WHO Member States) indicating that there is marketing authorization and use of these vaccines in these countries. WHO is working with Member States to support the introduction of seasonal influenza vaccines and expansion of national programmes to reduce annual morbidity and mortality from seasonal epidemics and support pandemic preparedness.

Reference was made to the PIP Framework and their negotiations for access to pre-pandemic vaccines.

Supply chains and logistics

Supply chain vulnerabilities were repeatedly emphasized as critical determinants of equitable access. For diagnostics, shortages of key materials, machines, and cold chain capacity were flagged as challenges. Furthermore, there could also cascade into delays in downstream allocation of antivirals and vaccines. For antivirals, stockpiling and prepositioning were expected to occur primarily at supra-national or regional levels. Visibility into what was stockpiled—finished product versus active pharmaceutical ingredients—was seen as important and it was flagged that in some settings, legislation may permit stockpiling only of finished formulations of antivirals. Cross-cutting considerations on stockpiling were noted – e.g. the management of storage capacities and shelf-life/expiration dates. Surge capacity models (participants referred to them as 'ramp-up') were mentioned, whereby a selection of manufacturers could develop the capacity to rapidly scale-up production in an emergency to help mitigate these constraints, but this remains to be developed.

Across the supply chain, infection prevention and control measures (e.g. gloves, masks, etc) tailored to the specific pathogen were recognized as necessary safeguards. Availability of pathogen-specific infection prevention and control guidance would influence how manufacturing, packaging, transport, and distribution activities could be implemented, particularly in contexts with limited workforce resilience. The discussion also concerned how panic and fear in communities may impede adherence to evidence-based infection prevention and control guidance and waste management.

Regional specificities

There is likely to be a range of operational challenges that countries throughout the six WHO regions will experience including delivery considerations, confidence in the available medical countermeasures, and potential lack of experience with available diagnostic tests. Given the diverse contexts, there will be regional specificities that can be addressed through dedicated technical and financial support.

Regional procurement platforms were anticipated to play an increasingly prominent role. Still, their capacities and functionality remain variable and there is a need for a greater understanding of the response capacities.

Summary of key aspects

- Vaccine production cycles carry inherent planning considerations and potential tensions between seasonal obligations and pandemic response needs.
- Visibility into stockpiles, including the form of products held (finished versus APIs), is needed for planning.
- Infection prevention and control requirements across the supply chain influence the feasibility and safety of operations.
- Regional coordination bodies are likely to act as critical enablers but may have different capacities.
- Allocation decisions will only be as effective as the supply systems that support them.
- The Pandemic Influenza MCM Operationalization Working Group will enable coordination around existing commitments for pandemic MCMs.

3.4 Financing and incentives

Financing mechanisms and incentive structures were anticipated to be central determinants of timely access. In the early stages of the simulation, when suspected influenza A(H5N1) cases were first reported, the ability to activate financing quickly was seen as shaping the speed of procurement and deployment of diagnostics, antivirals, and vaccines. Mechanisms such as donor envelopes, Gavi's Day Zero Financing Facility, and emergency funds were highlighted as potential enablers, but their effectiveness was anticipated to depend heavily on pre-agreed rules, clarity of access, and the speed at which funds can be made available and flow.

Financing as a strategic lever

As the scenario moved into regional expansion, differences in financing readiness became more pronounced. Participants expressed that countries with pre-committed or flexible resources may be better positioned to access MCMs more quickly, while those dependent on ad hoc funding risked delays that constrained coverage of high-risk populations. Financing was therefore not only seen as a technical enabler of procurement but as a strategic lever influencing the introduction of MCMs and therefore impacting subsequent allocation rounds. This highlights the close interdependence with supply and country readiness. WHO and partners are providing technical and financial support to countries (including provision of guidance, trainings, and tools through the PIP Framework Partnership Contributions) to build national capacity/readiness for pandemic emergency MCM introduction. Delays in funding flows were expected not only to affect procurement but also to cascade into logistical challenges such as customs clearance, cold chain management, and last-mile delivery of products. This underscored that financial preparedness needs to be aligned with operational readiness if allocation decisions are to translate into real-world impact.

Incentives for manufacturers

Ensuring surge production capacity emerged as a recurrent theme, some participants expressed that the global capacity for antivirals and vaccines may be considered theoretically sufficient, however the willingness and ability of manufacturers to redirect or expand production during a pandemic depended on predictable incentives. SMTA2 agreements under the PIP Framework were seen as one mechanism providing some certainty. Other access arrangements were also mentioned as potentially valuable for the inter-pandemic and emergence periods. Traditionally these include tiered pricing arrangements, pre-financing of production, and advance purchase commitments. Discussions also pointed to the potential of risk-sharing mechanisms that could support manufacturing decisions at the very earliest stages, when demand is uncertain and commercial risks are high. Investing in broader-spectrum products with applications beyond the pandemic context was also anticipated to help reduce risk exposure and sustain preparedness capacity.

Risk, costing, and the "no-regrets" approach

Participants emphasized that financing decisions in the earliest phase of an outbreak will inevitably carry uncertainties and potential sunk costs. A "no-regrets" framing was seen as important, with early investments considered preferable to delays that could widen inequities and prolong transmission. Scenario-based costing exercises were anticipated as a valuable tool for shaping financing envelopes, both to assess the scale of potential needs and to guide efficient allocation of limited resources.

Summary of key aspects

- Early financing triggers are decisive in shaping the timeliness of MCM access.
- Risk-sharing mechanisms and broader-spectrum investments may help reduce uncertainty in the earliest stages.
- Scenario-based costing exercises would be valuable for anticipating funding needs and ensuring efficiency.
- A degree of sunk costs may be unavoidable and should be approached with a "no-regrets" perspective.
- Alignment of financing streams across donors, emergency funds, and national budgets would influence efficiency and coherence of the overall response.

3.5 Country readiness and impact on demand

In the earliest injects, focused on the detection of influenza A(H5N1) cases, the ability of countries to absorb and operationalize available diagnostics, antivirals, and vaccines was seen as dependent on various factors including their health systems' capacities, pandemic preparedness efforts, financing of operational response, and community engagement. Even when global mechanisms enabled access, operational delays were expected in contexts lacking capacities. Participants proposed the further recognition and integration of seasonal influenza vaccination programmes as a key component of influenza and respiratory disease pandemic preparedness and response efforts. Seasonal influenza vaccination programmes provide a foundation which can be leveraged in the event of a pandemic emergency to enable rapid and efficient delivery and uptake of pandemic vaccines. This corresponds with the growing evidence base that countries with seasonal influenza vaccination programmes leveraged these capacities for faster roll-out of 2009 pandemic and COVID-19 pandemic vaccines.

Readiness data during the interpandemic period

Discussions underscored that allocation frameworks could only be effective if national systems were prepared to translate supply into use. Readiness databases, containing information on existing capacity and infrastructure to manufacture MCMs locally, regulatory maturity levels, country-specific import and export legislation, National Deployment and Vaccination Plans status, and broader legal frameworks governing movement of goods during emergencies, were anticipated to be valuable tools for early situational analysis. Such information could help understand access constraints and anticipate which countries would be able to deploy MCMs immediately and which might require additional support, or which products would be more suited for different settings.

Demand dynamics and public perception

As the scenario escalated to regional expansion, demand factors became more complex. Participants anticipated that vaccine hesitancy, misinformation, and uncertainty about disease severity or vaccine safety or vaccine effectiveness could create uneven uptake across and within countries. Risk communication and community engagement were highlighted as critical to balance perceived demand with actual supply, ensuring that scarce resources were both accepted and effectively deployed. In particular, attitudes and capacities of healthcare workers, prior patterns of seasonal vaccine uptake and for other vaccines targeting high risk groups (e.g. adults), and likely wastage rates were anticipated as inputs to help guide assessment of realistic demand projections.

Timing of demand signals

Operational discussions revealed the importance of aligning demand signals with product availability. Countries with strong surveillance systems and established pathways for requesting support were expected to issue early and precise requests for antivirals and diagnostics, thereby facilitating timely deployment. By contrast, countries with weaker planning systems or fragmented governance might lag, creating mismatches between epidemiological needs, expressed demand, and actual allocation. This highlighted the necessity of accounting for data gaps or lack of capacities to gather data in allocation frameworks accordingly, to avoid leaving countries in need underserved.

Summary of key aspects

- Country readiness is likely to determine how efficiently allocated MCMs can be deployed, with National Deployment and Vaccination Plans, governance frameworks, operational funding, and regulatory maturity as critical factors.
- Factors such as public perceptions, mis/dis-information, and community engagement including healthcare worker attitudes and epidemiologic trends, and severity will influence uptake of MCMs.
- Readiness databases, integrating regulatory, operational, and governance details, could support more accurate assessments of country coping capacities.
- Allocation approaches should consider potential mismatches between realistic demand, funded demand, and needs based on epidemiological assessment.

3.6 Coordination, partnerships and urgent needs

The progressive injects—from early detection of influenza A(H5N1) cases to a full pandemic declaration—highlighted how multi-level coordination challenges could directly shape the timeliness and equity of access and allocation and deployment of MCMs. Participants anticipated that clear roles, defined convening mechanisms, and structured communication pathways are needed to ensure efficient activities across diverse actors.

Global and regional integration

At the global level, there are different roles that span from global mandates and legal responsibilities. However, in addition to the usual global partners coordination considerations, discussions emphasized that active alignment with regional actors requires further considerations. Regional bodies were seen as increasingly important partners with increasing capacities – including for supporting research, procurement and deployment, providing support to member countries, among others. The anticipated agility of regional actors suggested that they could serve as first responders, particularly in the early stages of an outbreak, provided they have the capacities, the procured products and the mandates to undertake such actions. Participants anticipated that regional investments – such as those in the European Commission with Health Emergency Preparedness and Response Authority's investments in regional coordination of clinical trials, the European Union's MCM Strategy, and initiatives like EUFAB's reserved vaccine manufacturing capacity, could significantly impact both regional and global responses. Leveraging these capacities for global benefit would depend on transparent coordination between global and regional frameworks. Questions of regional solidarity were also raised, extending beyond regional bodies activities to secure access agreements enabled by regional manufacturing capacity, but also facilitate bilateral arrangements between countries in the respective regions (e.g. some that may be producing MCMs and others that would seek access). These arrangements were

expected to influence allocation dynamics, underscoring the need for visibility and calibration across both regional and global systems.

Evolving partnerships across outbreak stages

The simulation suggested that engagement of partners would vary depending on the stage of the outbreak. Research-focused stakeholders, for instance, were expected to play a more active role in early phases, on activities such as potential immunogen design, trial coordination, and regulatory engagement for unapproved products. Other partners might initially limit their role to monitoring early signals, preparing to mobilize once data that is relevant for their areas of focus or that would signal a higher risk, was available. This variation highlighted the importance of visibility across the partner landscape, ensuring that early efforts on research and data could seamlessly inform later stages of response and allocation. Frameworks such as the Emergency Response Framework and work conducted under the Health Emergency, Preparedness, Response, and Resilience Framework (HPER) were noted as examples of mechanisms that could be leveraged to facilitate multi-sectoral alignment.

Humanitarian and cross-sectoral considerations

Humanitarian contexts were recognized as requiring specific planning —with a potential need for having a humanitarian buffer or similar arrangements—given their unique access barriers and operational challenges. It was also emphasized that previous attempts at creating humanitarian buffers were not successful. - Building on lessons from previous responses, approaches tailored to these settings were anticipated to be necessary for ensuring that highly vulnerable populations are able to access MCMs. In addition, effective coordination was expected to depend on integrating expertise from humanitarian setting.

Timing and triggers for action

One recurring theme concerned the timing and triggers for action. Across multiple injects, delays were linked to reliance on formal event determinations—particularly the public health emergency of international concern —as the primary signal for financing, procurement, and contractual activation. Demand signals that surfaced earlier in the “regional spread” phase were more difficult to translate into concrete mobilization until a formal determination was made, limiting the scope for proactive deployment. However, it was also considered that individual countries may trigger advance purchase agreements based on their national contexts - and that this may have a ripple effect on manufacturers transitioning to production of pandemic products. Participants also reflected that triggers are often interpreted narrowly as financial decisions, such as releasing funds or activating contracts, while in reality a wider set of non-financial resources could be mobilized earlier. Mapping these across partners was seen as potentially useful to support early response efforts. The notion of “no-regret actions” was raised in this context, although the way such actions are defined and applied across different institutions remained unclear.

Diagnostic preparedness as an urgent need

Diagnostic capacity and preparedness emerged as another area of systemic vulnerability. While this refers to a specific product, the simulation made clear that the implications reverberate across planning for all other MCMs. During the initial detection inject, the limited availability of polymerase chain reaction reagents was considered to potentially create bottlenecks for confirming cases, which in turn slow decisions on the allocation of antivirals and vaccines. In addition, the regulatory approvals

that validate diagnostic tests (e.g. the WHO prequalification and emergency use listing) to ensure quality assays are critical.

Unlike vaccines and antivirals, diagnostics do not benefit from structured stockpiles or global commitment mechanisms, leaving gaps in preparedness activities. It was clarified that during a pandemic emergency, the WHO will provide a proficiency panel containing diluted pandemic virus strains to support diagnostic development, helping manufacturers to validate their tests.

Discussions drew attention to the uneven transition planning for polymerase chain reaction machines procured during COVID-19, raising questions about whether these assets would be maintained for future needs.

For influenza A(H5N1), preparedness would also require a One Health approach, given the importance of testing in both animal and human populations for sustainable, and transparent influenza virus detection and reporting in relevant settings (including farmed animals, live markets, abattoirs, domestic/companion animals, wildlife, and environment), addressing funding, social, political, economic, and legal considerations. Most countries currently have capacities to detect H5, supported by WHO's annual external quality assurance programmes, but diagnostic availability for both surveillance and clinical care was seen as a decisive and challenging factor for pandemic response.

Participants anticipated that advance agreements with manufacturers, especially those with WHO-prequalified products, could play a role in supporting both timely access and regionalized production capacity. Increasing demand for differential diagnostics, the absence of target product profiles for the different use cases, and the lack of surge capacity planning were identified as central challenges. Parallel considerations included the need for developing cheap, rapid, sensitive and specific diagnostic tests for influenza detection and subtyping. Including point of care (Reverse transcription polymerase chain reaction and rapid diagnostic) molecular multiplex platforms and the importance of ensuring access to supply (reagents and cartridges), as well as the potential value of prepositioning commercial kits to shorten deployment timelines. Ongoing initiatives, such as WHO's forthcoming Unity validation protocol for Pathogen X Ag-rapid diagnostic test, the Pathogen Readiness Index developed by the Foundation for Innovative New Diagnostics, and the PIP Secretariat's manufacturers diagnostics landscape analysis were discussed as examples of mechanisms that could help strengthen readiness.

Summary of key aspects

- Global and regional bodies should coordinate throughout the entire pandemic response and regional mechanisms could act as first responders and need structured linkages with global allocation processes (which will also help to avoid divergence of funding).
- Partner engagement would evolve with outbreak phases; however, it is important to have visibility from the outset on the evolving context (e.g. epi, MCM access, country readiness etc) to facilitate timely actions.
- Humanitarian settings require dedicated approaches, building on lessons from past crises.
- Reliance only on formal event determinations may delay actions in some areas; broader early triggers and "no-regret" options could enable faster responses.
- Across to early response products such as diagnostics and antivirals were presented as an area requiring urgent preparedness attention and activities.

4. Summary analysis and implications for access and allocation planning

The simulation provided a valuable lens for understanding the complexity of allocating limited MCMs under conditions of uncertainty. Rather than isolated technical challenges, access and allocation decisions emerged as embedded within a network of interdependent systems—encompassing evidence generation to inform policy recommendations, regulatory pathways, manufacturing, supply chains, financing, country-level readiness, and public trust. The exercise highlighted the need to consider allocation not as a series of rigid choices, but as a dynamic process shaped by multiple interacting factors, with varying consequences depending on timing, context, and available resources.

4.1 Allocation as system-wide, iterative processes

Allocation decisions for diagnostics, antivirals, and vaccines are closely interlinked. Early phases of the simulation highlighted perspectives on specific constraints that have cascading implications for example, limited diagnostic capacity, can reverberate across other areas, affecting identification of critical clinical aspects (e.g., onset of infection) and hence impact MCM prioritization and deployment. Conversely, flexibility in the use of certain MCMs, such as offering antivirals particularly in the early stage of the pandemic emergency when vaccine capacity is limited, can partially mitigate delayed allocation risks. These dynamics suggest that allocation frameworks may be most effective when conceived as iterative, adaptive systems rather than fixed rules. They need to allow for context-specific adjustments based on changing epidemiological, operational, and social factors. This necessitates the awareness and use of real-time monitoring platforms and systems. The importance of having an allocation model in place, explainable and publicly accessible, adaptable and easily deployable, was underscored.

4.2 Diagnostics as a critical, yet variable, enabler

Diagnostics are at the basis of access to countermeasures. Without accurate diagnostics that identify affected populations, no vaccination strategy (except mass vaccination) nor treatment strategy can be successful. While they emerged as a central determinant of both timeliness and equity, their role is not uniform across contexts. In early detection scenarios, insufficient diagnostic capacity and supporting platforms may complicate requests for antivirals and inhibit accurate demand forecasting for antivirals and vaccines. However, the degree to which diagnostic limitations constrain allocation will depend on national surveillance and testing policies and networks, pre-existing laboratory infrastructure, and availability of diagnostic tests. Efforts to support development of rapid, sensitive, affordable point-of-care diagnostic tests for detecting influenza virus and antiviral resistance, validation studies, pre-positioned kits, and One Health approaches are valuable, but their effectiveness may vary with local capacities and supply chain realities.

4.3 Regional mechanisms will impact global allocation efforts

Regional coordination bodies can facilitate solidarity, pooled procurement, and early response capacity, potentially smoothing gaps between global guidance and national implementation. Examples from the European region suggest that these mechanisms can enhance the speed and coherence of allocation decisions. Yet regional effectiveness is not uniform: capacities, mandates, and integration with national systems differ, and political or logistical constraints can influence outcomes. Global and national planners should recognize both the potential and the limitations of regional bodies, building strategies that allow for variable engagement and capacity across regions.

4.4 Financing: timing and flexibility over absolute availability

The simulation emphasized that the availability of funds is as consequential as the conditions under which they can be accessed. Early triggers, pre-committed envelopes, and risk-sharing mechanisms shape whether allocation strategies are proactive or reactive. Even when resources exist in principle, delays in activation or rigid contractual conditions may slow procurement, production scale-up, and distribution. Financial planning therefore benefits from flexibility and contingency planning, acknowledging that allocation decisions may need to adjust in response to evolving funding landscapes and operational realities.

4.5 Manufacturing, supply chains, and operational complexity

Production capacity, supply chain resilience, and operational readiness shape what allocation decisions are feasible in practice. Lead times for production of vaccines and antivirals, seasonal vaccine contractual obligations, stockpile forms (e.g. virtual, physical, etc), cold chain requirements, and regional surge capacities all influence the timeliness and equity of distribution. Yet these factors are not fixed constraints: adjustments—such as an informed, timely, and coordinated decision to switch to pandemic vaccine production, leveraging regional manufacturing platforms, or adapting stockpile composition – could broaden the scope of options, albeit mindful of the potential risk of trade-offs in risk, cost, or access to seasonal vaccine and other products.

4.6 Country readiness and nuanced demand considerations

National capacities to absorb and operationalize MCMs significantly influence the impact of allocation decisions, yet these capacities vary widely. Factors such as regulatory maturity, operational infrastructure, governance and legal frameworks, and availability of National Deployment and Vaccination Plans influence speed of deployment, but their relative importance differs across countries. Similarly, demand is shaped not only by epidemiologic need but also by social, behavioral, and political factors—including vaccine acceptance and demand, misinformation, and trust in authorities. Effective allocation frameworks should therefore incorporate both supply-side constraints and realistic demand assessments, recognizing that uptake may diverge from assessed needs at a given time. Readiness assessments, technical support packages, and integration with broader health system strengthening efforts are essential to ensure equitable and effective deployment.

5. Way forward

The simulation exercise served not only as a test of current allocation approaches but also as a bridge toward refining the systems that will serve updated approaches to global pandemic preparedness.

During the simulation exercise, participants' experience with access and allocation of medical countermeasures enabled productive discussions. In particular, the participants' insights on how previous global access and allocation activities were recognized as providing valuable learnings for the development of the Global Allocation Framework. Building on the lessons and gaps identified, three immediate pathways forward have been outlined.

5.1 Integration of insights into the Global Allocation Framework

Insights from the exercise will directly inform the ongoing revision of WHO's Global Allocation Framework for pandemic influenza MCMs. WHO will engage Member States through structured consultations to further refine and validate the proposed approaches, ensuring alignment with both global priorities and national capacities. Mechanisms for inclusive feedback and transparent decision-making will be embedded to ensure the framework is both responsive and representative.

5.2 Establishing a cross-cutting working group

A dedicated operational working group will be convened to address the most pressing cross-cutting issues identified in the simulation. Priorities include enhancing information-sharing, improving demand forecasting, clarifying frameworks for bilateral donations and explore visibility-driven mechanisms for advance purchase agreements, standardizing product-agnostic legal documentation, and advancing modelling and costing of MCM access, allocation, and deployment strategies. The group will draw on expertise from WHO, partner agencies, regional bodies, and other relevant stakeholders as appropriate.

5.3 Strengthening operational strategies

The exercise highlighted key areas where operational planning can be strengthened. These include clearer articulation of allocation triggers across products, development of templates for financing and contracting strategies to enable earlier action, and coordination mechanisms that account for the mandates of emerging regional bodies. Translating these insights into practical tools—such as allocation playbooks, guidance notes, and standard operating procedures—will be a central focus of pandemic preparedness efforts going forward. A monitoring and evaluation framework will be developed to track progress and impact.

Annexes

Annex 1. Agenda

Day 1: Setting the stage and initial response

08:30 - 09:00	Registration and welcome coffee
09:00 - 09:30	Opening remarks - *Dr. Chikwe Ihekweazu, Executive Director, WHO Health Emergencies Programme*
09:30 -10:00	Participants introductions
10:00 - 10:30	Presentation: Considerations for MCM access, allocation and deployment - WHO
10:30 - 13:00	Simulation sessions: Initial access approaches and allocation decisions including prioritization strategies for vaccine, diagnostics, and antiviral distribution *(Scenario will consider a severe outbreak – pandemic not declared)*

Sessions will be split into:

10:30-11:15	*Main scenario and inject 1*
11:15-12:15	*Inject 2*
12:15-13:00	*Inject 3*

Overall, the morning sessions will cover:

- Activation triggers for access mechanisms
- Overcoming challenges in access such as (but not only): production delays, capacity limitations, supplier dependencies, lack of stockpiling visibility of critical MCMs prior to the outbreak.

13:00 - 14:00	Lunch break
13:30 - 15:00	Continuation of scenario and simulation session – *containing Inject* 4
15:00 - 15:15	Coffee break
15:15 -16:15	Continuation of scenario and simulation session – *Session will contain escalation of main scenario to pandemic (new scenario) and inject 1*

Overall afternoon sessions will cover:

- Coordination of gathering of input data for allocation decisions – supply and demand data, variables for allocation (epi, equity-related, country readiness etc)
- Process pathways and timelines
- Information exchange on other potential allocation frameworks
- Scale-up of response with injects targeting specific product specificities

16:15 - 17:00	Buffer time for discussions and Day1 overall reflections

Day 2: Full pandemic response and activation of all response capabilities

09:00 - 10:30 Continuation of scenario and simulation session

Sessions will be split into:

09:00-10:00 *Inject 2*

10:00-11:05 *Inject 3*

11:05 - 11:25 Coffee Break

11:25 - 12:25 Continuation of scenario and simulation session – *containing Inject 4*

Overall, the morning sessions will cover:

- Scale-up of response with injects targeting specific product specificities
- New allocations considerations considering new changes
- Process pathways and timelines

12:25 - 13:30: Lunch Break

13:30 – 14:30: Continuation of scenario and simulation session – *containing Inject 5*

Overall afternoon session will cover:

- Communication challenges

14:30 - 15:00 Plenary discussion: Discussion based on the overall simulation exercise

15:00 - 15:15 Coffee Break

15:15 - 15:45 Plenary discussion cont: Discussion based on the overall simulation exercise

15:45 – 16:00 Closing remarks - *Dr Maria D. Van Kerkhove, Director, a.i. Department of Epidemic and Pandemic Management, WHO Health Emergencies Programme, WHO*

Annex 2. List of participants

Participant Name	Organization
Saul Walker Director, Public Partnerships	Coalition for Epidemic Preparedness Innovations (CEPI)
Roshni Best *(online)* Emergency Response Senior Manager	Coalition for Epidemic Preparedness Innovations (CEPI)
Rajinder Kumar Suri *(online)* Chief Executive Officer	Developing Countries Vaccine Manufacturers Network (DCVMN)
Emmanuel Agogo Director, Pandemic Threats	Foundation for Innovative New Diagnostics (FIND)
Hannah Kettler Head, Global Health Security	Gavi, the Vaccine Alliance
Sophie Mathewson Senior Manager, Global Health Security	Gavi, the Vaccine Alliance
Simon Allan Lead, Market Shaping Team, Global Health Security	Gavi, the Vaccine Alliance
Luciana Da Silva Santos Programme manager	Health Emergency Preparedness and Response (HERA)
Paula Barbosa Associate Director, Vaccines Policy	International Federation of Pharmaceutical Manufacturers and Associations (IFPMA)
Gwendolene Eamer Senior Officer, Public Health in Emergencies	International Federation of Red Cross and Red Crescent Societies (IFRC)

Participant Name	Organization
Ann Ottosen Senior Manager, Supply Division	United Nations Children's Fund (UNICEF)
Siri Bjornstad *(online)* procurement leader	United Nations Children's Fund (UNICEF)
Hans Christiansen *(online)* Supply Division's Vaccine Centre	United Nations Children's Fund (UNICEF)
Johannes Kraus *(online)* Coordinator, Vaccine Center.	United Nations Children's Fund (UNICEF)
Nagwa Hasanin Senior advisor	United Nations Children's Fund (UNICEF)
Anu Puri Consultant, SBC/Demand Promotion for Vaccine Uptake Immunization Division	United Nations Children's Fund (UNICEF)
Maria Wilda Silva Immunization Specialist	United Nations Children's Fund (UNICEF)
Carmen Perez Casas Technical Manager	UNITAID
Pearl Ngugi Technical Officer	UNITAID
Tiwadayo Braimoh Policy Advocacy	Medicines Patent Pool (MPP)
WHO REGIONAL OFFICES	
Reena Hemendra Medical Officer (Immunization in Emergencies)	Regional Office for Africa
Francesco Nogareda Special Program Comprehensive Immunization (CIM)	Regional Office for the Americas/Pan American Sanitary Bureau
Marisol Valenzuela Lara Emergency Programme Technical Officer	Regional Office for the Americas/Pan American Sanitary Bureau

Participant Name	Organization
Pernille Jorgensen Technical Officer	Regional Office for Europe
Emanuele Bruni Programme Area Manager (Emergency Operations)	Regional Office for Europe
Pushpa Wijesinghe *(online)* Programme Area Manager New Delhi	Regional Office for South-East Asia
Misheel Enkhdalai Consultant	Regional Office for the Western Pacific
WHO HEADQUARTERS	*WHO Departments or Units*
Chikwe Ihekweazu Executive Director	WHO Health Emergencies Programme
Maria Van Kerkhove Director, a.i. Department of Epidemic and Pandemic Management	Department of Epidemic and Pandemic Management
Tim Nguyen Unit Head, Medical Countermeasures Unit	Medical Countermeasures Unit
Ioana Ghiga Technical Officer	Medical Countermeasures Unit
Alexandra Hill (rapporteur) Technical Officer	Medical Countermeasures Unit
Daniel Eibach (rapporteur) Technical Officer	Medical Countermeasures Unit
Zerthun Alemu Assistant To Unit Head	Medical Countermeasures Unit
Wenqing Zhang Unit Head, Global Influenza Programme	Global Influenza Programme Unit
Sandra Jackson *(online)* Technical Officer, Global Influenza Programme	Global Influenza Programme Unit

Participant Name	Organization
Anne Huvos Unit Head, Pandemic Influenza Preparedness	Pandemic Influenza Preparedness Unit
Olga Kim Technical Officer	Pandemic Influenza Preparedness Unit
Alexandra Kontic *(online)* Consultant	Pandemic Influenza Preparedness Unit
Kate Rawlings Technical Officer	Pandemic Influenza Preparedness Unit
Lorenzo Subissi Technical Officer	Emerging Diseases and Zoonoses Unit
Jamie Rylance Case Management Expert	Health Care Readiness Unit
Alireza Khadem Broojerdi Team Lead, Regulatory Systems Strengthening	Regulation and Prequalification Department
Razieh Ostad Scientist, Regulatory Systems Strengthening	Regulation and Prequalification Department
Tara Lavanya Prasad Team Lead, Global Access	Agenda, Policy and Strategy Unit
Georgios Stathopoulos Technical Officer, (Vaccine Markets)	Agenda, Policy and Strategy Unit
Alba Vilajeliu Technical Officer	Expanded Programme on Immunization Unit

World Health Organization
20 Avenue Appia
CH-1211 Geneva 27
Switzerland